ASSIGNMENT - 2

SOLUTIONS

Q1 (a) As the Gini coefficient gets higher, that means a country's income inequality increases, this leads to the detriment of a country's sociopolitical stability.

So for every tenth of a unit increase (0.10) in the Gini coefficient, we would be likely seeing a 3.32 unit increase in a country's sociopolitical stability.

(b) There is a 0.3 increase in the Gini coefficient, so the increase in SPI will be 33.2 (0.3) which is 9.96. This will result in greater socio-political instability.

(c) Using the standard t-test $t = \frac{33.2}{11.8} = 2.8136$ for testing the null hypothesis that the slope coefficient is 0. For 38 degrees of freedom between 2.021 and 2.042 (using a two sided test), so the estimated slope is statistically significant at the 5% level.

(d) Based on the regression results, we can conclude there is a positive relationship between higher income inequality and greater political instability, although we cannot make a causal statement about the relationship.

Q 2 (a) The elasticity is -1.34. It is significantly different from zero, for the t-value under the null hypothesis that the true elasticity coefficient of zero is: $t = \frac{-1.34}{0.32}$

$= -4.1875$

The p value of obtaining such a t value is extremely low.

However, the elasticity coefficient is ~~not~~ different from one because under the null hypothesis that the true elasticity is 1, the t-value is :

$$t = \frac{-1.34-1}{0.32} = \cancel{-1.062} \; -7.3125$$

This t-value is significantly different from one.

(b) The income elasticity of demand is 0.17.

$$t = \frac{0.17}{0.20} = 0.85$$

The critical t-value is 2.021 which exceeds the calculated t-value. Hence it is significantly not different from zero.

(c) $$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k}$$

Hence $\bar{R}^2 = 0.27$,

$n = 46$

$k = 3$

$R^2 = 0.3025$

Q 3) The regression results are:

$\hat{\alpha}_1 = -3.00$ $\hat{\alpha}_2 = 3.50$

$\hat{\lambda}_1 = 4.00$ $\hat{\lambda}_2 = -1.357$

$\hat{\beta}_1 = 2.00$ $\hat{\beta}_2 = 1.00$ $\hat{\beta}_3 = -1.00$

(a) No given model (3) is the true model, $\hat{\alpha}_2$ is a biased estimator of β_2.

(b) No. $\hat{\lambda}_3$ is a biased estimator of β_3 for the same above reason.

We conclude that mis-specifying an equation can lead to biased estimation of the parameters of the true model.